

Xinyuan Yan

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Research Interests

Visual Analytics; Explainable AI; Topological Data Analysis; Human-AI Interaction; Information Visualization

Education

University of Utah, Salt Lake City, USA 08/2021 – present
Ph.D. in Computer Science (Expected Summer 2026) GPA: 3.97/4.0
Advisor: Bei Wang Phillips
Committee: Vivek Srikumar, Andrew McNutt, Shusen Liu, Mennatallah El-Assady
Shandong University, Qingdao, China 09/2015 – 06/2019
Bachelor of Engineering in Computer Science and Technology GPA: 3.69/4.0

Work Experience

- **Research Assistant**, SCI Institute, University of Utah 09/2021 – present
- **Visiting PhD Student**, IVIA Lab, ETH Zurich 08/2025 – 09/2025
- **Research Intern**, Bosch Center for Artificial Intelligence, USA 05/2023 – 08/2023
- **Research Assistant**, Department of Statistics, Southern University of Science and Technology 09/2020 – 07/2021
- **Research Assistant**, School of Computer Science and Technology, Shandong University 07/2018 – 05/2020

Publications and Manuscripts

1. **Explainable Mapper: Charting LLM Embedding Spaces Using Perturbation-Based Explanation and Verification Agents.** Xinyuan Yan, Rita Sevastjanova, Sinie van der Ben, Mennatallah El-Assady, Bei Wang. arXiv:2507.18607, 2025.
2. **Visual Exploration of Feature Relationships in Sparse Autoencoders with Curated Concepts.** Xinyuan Yan, Shusen Liu, Kowshik Thopalli, Bei Wang. *Mechanistic Interpretability Workshop at NeurIPS, 2025*. arXiv:2511.06048.
3. **VISLIX: An XAI Framework for Validating Vision Models with Slice Discovery and Analysis.** Xinyuan Yan, Xiwei Xuan, Jorge H. Piazzentin Ono, Jiajing Guo, Vikram Mohanty, Shekar Arvind Kumar, Liang Gou, Bei Wang, Liu Ren. *Eurographics Conference on Visualization (EuroVis), Computer Graphics Forum, 44(3), e70125, 2025*.
4. **Flexible Topology Tracking with Partial Optimal Transport.** Mingzhe Li, Xinyuan Yan, Lin Yan, Tom Needham, Bei Wang. *IEEE Transactions on Visualization and Computer Graphics (TVCG), 31(1), 7951-7969, 2025*.
5. **Teaching People LLM's Errors and Getting it Right.** Nathan Stringham, Fateme Hashemi Chaleshtori, Xinyuan Yan, Zhichao Xu, Bei Wang, Ana Marasovic. arXiv:2512.21422, 2025.
6. **See or Recall: A Sanity Check for the Role of Vision in Solving Visualization Question Answer Tasks with Multimodal LLMs.** Zhimin Li, Haichao Miao, Xinyuan Yan, Valerio Pascucci, Matthew Berger, Shusen Liu. arXiv:2504.09809, 2025.
7. **EulerMerge: Simplifying Euler Diagrams Through Set Merges.** Xinyuan Yan, Peter Rodgers, Peter Rottmann, Daniel Archambault, Jan-Henrik Haunert, Bei Wang. *Proceedings of the 14th International Conference on the Theory and Application of Diagrams (DIAGRAMS), 2024*.
8. **Exploring Visualization for Fairness in AI Education.** Xinyuan Yan, Youjia Zhou, Arul Mishra, Himanshu Mishra, Bei Wang. *IEEE 17th Pacific Visualization Conference (PacificVis), 2024*.
9. **Generating Euler Diagrams Through Combinatorial Optimization.** Peter Rottmann, Peter Rodgers, Xinyuan Yan, Daniel Archambault, Bei Wang, Jan-Henrik Haunert. *Eurographics Conference on Visualization (EuroVis), Computer Graphics Forum, 43(3), e15089, 2024*.
10. **Enable Decision Making for Battery Electric Bus Deployment Using Robust High-Resolution Interdependent Visualization.** Gabrielius A. Kudirka, Xinyuan Yan, Sarah Kunzler, Yirong Zhou, Bei Wang, Xiaoyue Cathy Liu.

Transportation Research Board (TRB) 102nd Annual Meeting, 2023.

11. **Turing Galaxy: Visualizing Turing Award Laureates.** Xinyuan Yan, Yifang Ma. *IEEE Information Visualization Conference (VIS) Poster, 2021.*
12. **Elastic Tree Layouts for Interactive Exploration of Mentorship.** Xinyuan Yan, Yifang Ma. *Eurographics Conference on Visualization (EuroVis) Poster, 2021. (EuroVis 2021 Best Poster Award)*
13. **Interactive Structure-aware Blending of Diverse Edge Bundling Visualizations.** Yunhai Wang, Mingliang Xue, Yanyan Wang, Xinyuan Yan, Baoquan Chen, Chi-Wing Fu, Christophe Hurter. *IEEE Transactions on Visualization and Computer Graphics (Proceedings of InfoVIS), 2019.*

Selected Projects

1. Project Lead, **TopoAlign: A Visual Analytics Framework for Topological Comparison of Model Representations**

Collaborators: Bei Wang, Rita Sevastjanova, Menna El-Assady

12/2025 – present

Description: We are developing a framework for model comparison by constructing mapper graphs over shared data to compare the topological structures of models' latent spaces. By tracking changes in topological features (e.g., nodes, paths, connected components), the system reveals similarities and differences in how models organize knowledge, supporting expert-driven interpretation and model selection.

2. Project Lead, **Visual Exploration of Feature Relationships in Sparse Autoencoders with Curated Concepts**

Collaborators: Shusen Liu, Kowshik Thopalli, Bei Wang

05/2025 – 09/2025

Description: We present SAE Explorer, a visual analytics system that combines UMAP and mapper graphs, a topological technique, to analyze relationships among LLM features learned by Sparse Autoencoders (SAEs). We focus on human-understandable features and support feature comparisons across SAEs and model layers.

3. Project Lead, **Explainable Mapper: Charting LLM Embedding Spaces Using Perturbation-Based Explanation and Verification Agents**

Collaborators: Bei Wang, Rita Sevastjanova, Sinie van der Ben, Menna El-Assady

11/2024 – 09/2025

Description: We present the Explainable Mapper workspace, which uses mapper graphs to analyze the topological structure of LLM embedding spaces. It introduces LLM-based Explainer and Verifier agents that generate and validate explanations of different graph elements regarding the encoded linguistic properties, through operations of summarization, comparison, and perturbation.

4. Project Lead, **VISLIX: An XAI Framework for Validating Vision Models with Slice Discovery and Analysis**

Collaborators: Jorge H. Piazzentin Ono, Xiwei Xuan, Liang Gou, Jiajing Guo, Vikram Mohanty, Bei Wang 05/2023 – 12/2024

Description: Leveraging large language models and vision-language models, we developed a visual analytics framework that automatically identifies and explains systematic errors (i.e., data slices) in vision models, and empowers experts to efficiently explore slices and interactively test hypothetical slices.

5. Project Lead, **EulerMerge: Simplifying Euler Diagrams Through Set Merges**

01/2023 – 11/2023

Collaborators: Peter Rodgers, Peter Rottmann, Daniel Archambault, Jan-Henrik Haunert, Bei Wang

Description: We proposed a topology-based method for drawing scalable and readable Euler diagrams through set merging operations.

6. Project Lead, **Exploring Visualization for Fairness in AI Education**

Collaborators: Bei Wang, Youjia Zhou, Arul Mishra, Himanshu Mishra

01/2022 – 03/2023

Description: We developed interactive visualizations to educate non-technical students about fairness in AI, covering how a logistic regression model works and how to detect, quantify, and mitigate bias. We conducted a user study with 413 participants to evaluate learning outcomes. The module has been used in six courses at the University of Utah.

Teaching

Guest Lecturer, CS 3090: Ethics in Computing, University of Utah

Fall 2025

Guest Lecturer, CS 3090: Ethics in Computing, University of Utah

Fall 2024

Teaching Assistant, CS 3960: Algorithm Fairness in Machine Learning, University of Utah

Spring 2023

Teaching Assistant, CS/DS 3190: Foundations of Data Analysis, University of Utah

Fall 2022

Talks

Eurographics Conference on Visualization , Luxembourg <i>VISLIX: An XAI Framework for Validating Vision Models with Slice Discovery and Analysis</i>	06/2025
International Conference on the Theory and Application of Diagrams , Münster, Germany <i>EulerMerge: Simplifying Euler Diagrams Through Set Merges</i> <i>Fostering Responsible AI through Visual Analytics</i> (Graduate Symposium)	10/2024
IEEE Pacific Visualization Conference , Tokyo, Japan <i>Exploring Visualization for Fairness in AI Education</i>	04/2024
Bias, Ethics, and Trustworthiness in Data Science and AI , University of Utah <i>Exploring Visualization for Fairness in AI Education</i>	09/2023
Eurographics Conference on Visualization <i>Elastic Tree Layouts for Interactive Exploration of Mentorship</i>	06/2021
IEEE VIS Conference , Vancouver, Canada <i>Interactive Structure-aware Blending of Diverse Edge Bundling Visualization</i>	10/2019

Awards and Honors

EuroVis 2021 Best Poster Award	2021
Third Prize, Data Visualization Contest of Young Researchers of the Complex System Society	2020
Outstanding Undergraduate Thesis, Shandong University (Top 2%)	2019
Meritorious Winner, Mathematical Contest in Modeling (International, Top 7%)	2017

Skills

Python, R, JavaScript, React.js, D3.js, Flask, PyTorch, Hugging Face, Transformers, Git, Linux